Climate-risk compliance evidence, built on Earth Observation and AI

The physical climate-risk evidence layer for CSRD / ESRS E1, EU Taxonomy DNSH and EUDR — audit-ready, transparently priced, and honest about uncertainty.

Prepared for the Luxembourg Space Agency · September 2026

Motaz Omarien, Founder · motaz3d@gmail.com · +352 661811680 · talaix.com

THE PROBLEM

Mandatory climate disclosure meets a missing evidence layer

Physical climate risk is intensifying

Documented events in our own loss registry illustrate the scale: the July 2021 Western European floods, the 2002 Elbe/Danube floods and the 2018 Attica wildfires — multi-billion-euro events hitting assets that were never screened for hazard exposure.

Disclosure is now law, on a fixed calendar

CSRD/ESRS E1, EU Taxonomy DNSH and EUDR force companies to disclose **site-level physical climate-risk evidence** — acute and chronic hazards at the coordinates of material assets, with stated evidence status and limitations.

FY2024 →
CSRD Wave 1 reports already published — the calendar is running

30 Dec 2025
EUDR in force for large operators & traders

~12–15K
EU companies in post-Omnibus CSRD scope (est.)

5–7 figures/yr
typical contract at specialist climate-risk vendors

The gap: ESG/CSRD reporting suites (Workiva, Position Green, SAP, IBM…) manage workflow and carbon accounting — **none generates physical climate-risk evidence**. The specialists who do (Jupiter, Climate X, Mitiga, XDI…) sell opaque five-to-seven-figure annual enterprise contracts. Mid-market companies and their auditors are left with consultants or nothing.

What the market is missing

- **Audit-ready evidence** — sourced, reproducible, with an honest evidence status per datapoint · **Transparent pricing** — per-report and per-month, not six-figure contracts · **Global coverage** — EU companies today; non-EU exporters (EUDR, CSRD Wave 4) tomorrow

Talaix: site coordinates in, audit-ready evidence out

Talaix converts Earth Observation, environmental modelling and documented loss data into **physical climate-risk evidence packs** that plug directly into CSRD/ESRS E1, EU Taxonomy DNSH and EUDR workflows.

1 · Locate
Customer provides site coordinates of material assets

2 · Screen
Multi-hazard engine screens 8 hazards + exposure at those coordinates

3 · Evidence
Each datapoint tagged OBSERVED → ESTIMATED → FRAMEWORK → UNKNOWN

4 · Report
Audit-ready pack: ESRS E1 mapping + XBRL, documented losses, limitations

Why customers buy

- **The budget already exists** — compliance spend is mandatory and deadline-driven, not discretionary
- **Price transparency as a wedge** — €19–39 per report, €49–249/month tiers, published on the website
- **Near-zero delivery cost** — automated engine, no consultants; marginal cost per customer ≈ 0
- **Self-serve funnel, live today** — free CSRD scope check → €19–39 sample pack on the customer's own sites → €490 pilot → subscription

The trust anchor

Honesty contract: "unavailable is stated, never filled in." Every figure carries its source, method, reference period and evidence class. The analytical engine is open-source (**tore**, EUPL-1.2) with a public source registry — verifiable by any auditor, not a black box.

€19	≈ €0	8	100%
entry price — evidence on the customer's own sites	marginal cost per generated report	hazards screened per evidence pack	of outputs carry source + evidence class

TECHNOLOGY

Deep-tech core: EO + environmental modelling + compliance rules-as-data

Earth Observation & data

- Copernicus Sentinel-1/2/3, EFFIS, C3S climate services
- NASA FIRMS observed-fire history layer
- OSM/ohsome exposure (buildings, critical facilities, infrastructure)
- Luxembourg ACT cadastre — real building floor areas; Eurostat construction-cost calibration

TX Engine — multi-hazard core

- 8 hazards: wildfire, flood, drought, heat, wind, coastal, cyclone, earthquake
- FWI, fire-spread, ignition and smoke modelling; soil-to-fuel moisture transfer
- Exposure & economic screening with declared-assumption discipline
- Documented-loss registry (officially sourced figures only)

CsrdTX — compliance layer

- ESRS E1 datapoint mapping as versioned rules-as-data (5 JSON registries)
- XBRL output + /api/v2/csrd (shipped 2026-09-05)
- Scenario / time-horizon fields, E1-9 financial-effects context
- Evidence classes + public source registry on every output

Delivery surface: Flask API v2 (live), PDF report generation, Python & JS SDKs, QGIS plugin, self-serve web funnel with Stripe billing. Engine mirrored open-source at **tore** (EUPL-1.2).

Quality bar: 1,670 automated tests across 104 test files; benchmark suite and validation framework against documented events; screening-level labels until validation completes.

Ground truth (Stage 2): a soil- and fuel-moisture sensor testbed is planned for field validation — bridging the EO models to measured ground truth with hardware in the loop.

DEVELOPMENT STAGE

Working platform, pre-revenue, pilot-ready

8	1,670	2026-09-05	Live
climate hazards modelled by the TX engine	automated tests across 104 test files	CsrdTX compliance engine + XBRL + API shipped	pricing + self-serve funnel + Stripe billing

What exists today (not slideware)

- Multi-hazard engine, API v2, PDF evidence reports, SDKs, QGIS plugin — all operational
- Compliance products: ESRS E1 evidence brief + published case study, scope checker, pricing page
- Open-source engine mirror (**tore**, EUPL-1.2) published for programme co-funding applications (NLnet Restack, ESA BIC Luxembourg)
- Sales funnel live; first-customer motion ready (scope check → sample → pilot → subscription)

Honest status

- **Pre-revenue** — zero paying customers today; pilot programme opens the first wave
- Screening-level labels until model validation vs documented events completes
- Company formation: Luxembourg S.à r.l. — planned with this round, with LSA guidance on the fitting instrument (ESA BIC Luxembourg / LuxIMPULSE)
- Founder-built to date — this round funds the first team (see Team)

Roadmap to seed

Q4 2026
Luxembourg S.à r.l. incorporated; IP assigned to the company; trademark filed (EUIPO); ESA BIC Luxembourg application submitted

H1 2027
Model validation vs documented events; sensor testbed live; first paying pilots

H2 2027
Pilots → subscriptions; first GRC-suite integration; EUDR exporter wave

2028
Seed round; EIC Accelerator application (€2.5M grant + equity) with pilot evidence

MARKET OPPORTUNITY

A regulated-demand market with a published calendar

Regulation	Population	First reports	Status
CSRD Wave 1 (large PIEs)	largest EU companies	FY2024 (published 2025)	In force
CSRD Wave 2 (other large)	~12,000–15,000 post-Omnibus (est.)	FY2027 (published 2028)	In force; +2y delay
CSRD Wave 4 (non-EU groups)	non-EU >€150M EU turnover	FY2028 (published 2029)	In force
EUDR operators & traders	EU + global commodity exporters	30 Dec 2025 / 30 Jun 2026	In force now
EU Taxonomy DNSH + CSDDD	large undertakings	2027–2028	In force

Bottom-up sizing (our honesty rules apply)

- **Beachhead:** EU mid-market companies in CSRD scope that cannot access enterprise vendors — unserved at transparent price points
- **Least-contested niche:** non-EU exporters (EUDR geolocation screening, CSRD Wave 4) — regulation in force, evidence is satellite-native
- **Channel multiplier:** 17 ESG/CSRD suites need a physical-risk evidence layer — integration, not competition

Demand validation

- Incumbents are buying this layer: MSCI acquired First Street (~$120M); ISS STOXX acquired Sust Global; Moody's acquired RMS ($2B) — physical-risk evidence is a must-have
- Consolidation at the top leaves the mid-market open — our entry point
- Post-Omnibus scope figures are estimates and presented as such (claims discipline is our brand)

BUSINESS MODEL

Self-serve SaaS + evidence packs + partner API

Product	Price (published)	Buyer / motion
Decision report (per site)	€19 one-time	Self-serve sample on customer's own sites
Scientific report (per site)	€39 one-time	Deeper evidence pack for auditors
Professional	€49 / month (€490 / yr)	Sustainability & risk officers
Business + seats	€249 / month + €25 / seat	Multi-site companies, consultancies
Pilot programme	€490 fixed	3-month structured pilot → publishable case study
Partner API (planned)	usage-based	GRC suites & auditors embedding the evidence layer

Why this converts: the trigger is a legal deadline, the entry price is €19, and every step is self-serve. No sales team needed for the beachhead — outreach plus the free scope checker drives the funnel.

Why this scales: automated engine → marginal cost per report ≈ €0; the same evidence is reused by insurers (EIOPA/Solvency II) and banks (EBA Pillar 3 ESG) — expansion without new product lines.

COMPETITION

The gap: cheap, auditable, standalone physical-risk evidence

Category	Players	Physical-risk evidence?	Gap vs Talaix
ESG/CSRD suites (17 profiled)	Workiva, Position Green, IBM Envizi, SAP, Greenly, osapiens…	No (workflow + carbon)	Integration targets, not competitors
Climate-risk specialists (16 profiled)	Jupiter, Climate X, Mitiga, repath, XDI, First Street/MSCI…	Yes — bundled, opaque	5–7-figure contracts; no list prices; SME unserved
Open research models	CLIMADA (ETH Zurich)	Yes — research-grade	No compliance workflow, audit trail, or API product
Talaix	—	Yes — standalone packs	€19–249 transparent pricing, evidence classes, open engine

Defensibility & the honest risk

- **Moat:** evidence-class discipline + public source registry + open engine = the trust anchor auditors respond to; curated loss/benchmark registries accumulate as proprietary data assets
- **Stated risk (we name it):** Cervest collapsed in 2024 selling cheap discretionary analytics. Our difference: mandatory-deadline pull (CSRD/EUDR) and near-zero delivery cost — engine, not consultants

INTELLECTUAL PROPERTY

Open-core IP strategy, clean chain of title into the Luxembourg company

Chain of title at incorporation

- **Contribution in kind (apport en nature):** the full pre-incorporation IP portfolio — code, models, knowledge registries, documentation, brand, domains — contributed to the S.à r.l. in exchange for founder shares (apport en nature), described in the incorporation deed
- **Tax-clean & VC-standard:** no immediate income-tax charge on a genuine contribution in kind, supported by a contemporaneous valuation memo; no licence/terbeschikking construction
- **Trademark:** "Talaix" to be filed at EUIPO (classes 9 & 42) at incorporation, held by the company
- **Contributors & team:** CLA to govern the open-source repo (company keeps enforce/relicence rights); every hire signs explicit IP assignment

Open core — why the moat holds

- **Open (tore, EUPL-1.2):** the analytical engine — auditor trust anchor, public-funding eligibility (ESA BIC Luxembourg / LuxIMPULSE pipeline); interoperable copyleft does not contaminate the API-calling proprietary layer and blocks competitors from enclosing the engine
- **Closed (the company):** web platform, accounts & billing, compliance knowledge registries (CSRD rules, loss registry, benchmarks, cascading graph), operational pipeline, brand
- **Precedent:** GitLab, Elastic, Confluent, QuestDB — venture-scale on open cores
- **LU IP economics:** R&D tax credit and IP box regime (≈5% effective rate on qualifying IP income) filed at incorporation → unlocks preferential taxation on qualifying IP profits
- **Patent assessment** planned for the soil-to-fuel moisture-transfer methods during field validation

What an investor gets: a company holding the brand, the commercial platform, the compliance data assets and the customer relationships — with the open engine as a credibility asset, not a leakage risk (EUPL-1.2 copyleft keeps commercial forks honest).

TEAM

Founder-led, building the Luxembourg core team

Motaz Omarien — Founder
Designed and built the entire platform single-handedly: the multi-hazard TX engine, EO ingestion, the CsrdTX compliance layer (rules-as-data + XBRL), API v2, billing and the self-serve funnel — 1,670 tests of engineering depth, plus the market study and compliance strategy behind the positioning. Based in Luxembourg; incorporating in Luxembourg.

Hiring plan for this round (2–3 FTE)

- **Climate/ML validation scientist** — model validation vs documented events (LIST / SnT / University of Luxembourg pool)
- **Geospatial software engineer** — EO pipeline depth, performance, partner API
- **Commercial & pilot lead** — pilot conversions, GRC-suite partnerships, EUDR exporter outreach

Advisory & ecosystem: building in the Luxembourg space-data ecosystem — LIST and SnT (University of Luxembourg) for scientific validation, Luxinnovation for company formation and grants; LSA guidance on the right programme tracks.

Raising €850K pre-seed — 24 months to seed-ready

The ask

- **€850K pre-seed** (equity or convertible) — complemented by Luxembourg/ESA programme co-funding (ESA BIC Luxembourg, LuxIMPULSE, ESA Business Applications feasibility study)
- 24 months of runway at a lean 3–4 FTE burn (~€25–35K/month) in Luxembourg
- **Calibration:** comparable European climate/EO early rounds closed at €1M–€1.8M (Dryad, repath, Mitiga, Coolset) — we ask less because the platform is already built
- **Non-dilutive stack extends runway:** ESA BIC Luxembourg / LuxIMPULSE, ESA Business Applications feasibility study, CASSINI (€100K)
- Company formation: Luxembourg S.à r.l.; founder already based in Luxembourg

Use of funds

Area	Share	What it buys
Product & validation	~55%	2 technical FTE; model validation campaign; E1-9 depth; partner API
Commercial	~25%	pilot lead; 3–5 pilots; GRC-suite + EUDR exporter outreach
Operations	~20%	Company setup, IP assignment, trademarks, hosting, compliance

Milestones this round buys (24 months)

- **M6:** S.à r.l. incorporated with clean IP chain; ESA BIC Luxembourg application filed; validation vs documented events published → screening labels upgraded where the evidence supports it
- **M12:** 3–5 paying pilots converted to subscriptions; first GRC-suite integration live; €50K+ ARR run-rate
- **M18:** EUDR exporter wave + EBA/EIOPA reuse pack; €150K ARR run-rate
- **M24:** seed-ready: €250K ARR trajectory, 2+ suite channels, EIC Accelerator application (€2.5M grant + equity) filed with pilot evidence

An EO-native platform: satellite data is not an input we buy — it is what we are b

Fit with the LSA opportunity

- Earth observation is the engine's foundation: Sentinel-2, Landsat, NASA FIRMS, ESA WorldCover and ~25 integrated open datasets
- EU disclosure law (CSRD/ESRS E1, EUDR) turns Earth observation into a deadline-driven paid market
- Programme co-funding (ESA BIC, LuxIMPULSE, ESA Business Applications) de-risks the €850K pre-seed round
- the commitment to Luxembourg is already made

Why Luxembourg

- National space strategy and a dense space-data ecosystem (LIST, SnT, Luxinnovation)
- ESA BIC Luxembourg / LuxIMPULSE as the company-formation ladder for a pre-company project
- Founder already lives in Luxembourg — incorporation here is the natural path
- English-speaking, business-friendly gateway to EU programmes and ESA channels

Next step: we would welcome a 30-minute introductory call to walk through the engine live (talaix.com — the product, not a demo video) and discuss fit with the Luxembourg Space Agency's programme strategy.

Motaz Omarien · Founder · motaz3d@gmail.com · +352 661811680 · talaix.com